

d-CCSSBU L and S Charts and Poisson u Chart Code and Their Outputs for Equipment A and B

1 Read Excel Data

Each Excel Data contains three spreadsheet: DAR, CAR, and Mic. Each spreadsheet should have variables Event, Location, Result, USL, LOD, LOQ (For Mic, it does not have LOD and LOQ). You need to change the following path to yours and download these data, code to your local system. You need to have R/RStudio with necessary R packages installed.

You can also realize the algorithms in the manuscript directly in other programming languages.

2 d-CCSSBU-L Function

The function `d_ccssbu_l()` uses Excel data as above to generate D-CCSSBU-L chart.

3 Selection of d for d-CCSSBU-L Function

- `d_ccssbu_l()` constructs the d-CCSSBU-L chart for a given value of d which is the first highest risk d clusters used for UCL and CL estimation.
- `make_d_ccssbu_l_arl1_wide_tbl()` is a helper function for `d_ccssbu_l_d_selection_arl1()`.
- `d_ccssbu_l_d_selection_arl1()` applies `d_ccssbu_l()` across all candidate values $d = 1, \dots, q$ using Excel input data and summarizes OOC counts and ARL_1 performance for DAR, CAR, and Mic residue types and use `make_d_ccssbu_l_arl1_wide_tbl()` to make an output table.
- `render_d_ccssbu_l_arl1_tbl()` renders the output table from `make_d_ccssbu_l_arl1_wide_tbl()` as a responsive tibble that automatically adjusts to page width.

- `render_d_ccssbu_l_recommendation_tbl()` generates the recommended d selection table for each residue type (DAR, CAR, and Mic).
- `plot_d_ccssbu_l_arl1_curves()` plot all ARL_1 as y-axis values and grid of k as x-axis for all d-CCSSBU-L charts and X-bar chart.

3.0.1 d-CCSSBU-L Selection of d and ARL_1 Results for Equipment A

Table 1: Recommended d values for the d-CCSSBU-L chart for Equipment A.

Residue	Recommended d	q	Decision case	Observed-behavior case	ARL1 case	A	U	W(d)	G(d)	Selection rule
DAR	1	3	Case 2	D1 intersection D2	XO	{1, 2, 3}	Empty	2.2649	1.2633	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
CAR	1	1	Case 2	D1 intersection D2	XO	{1}	Empty	33.2108	5.3344	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
Mic	1	1	Case 2	D1 intersection D2	XO	{1}	Empty	3.6720	0.8032	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains

Figure 1 visualizes the ARL_1 curves for Equipment A. Four observations can be made:

- For each candidate d-CCSSBU-L chart, ARL_1 decreases as the additive location shift magnitude k increases, indicating improved OOC detection performance for larger process shifts.
- The ARL_1 curves differ substantially across d values, demonstrating the importance of optimizing d . For DAR, the candidate with $d = 1$ consistently achieves lower ARL_1 values than the candidates with $d = 2$ and $d = 3$, indicating superior OOC sensitivity. These differences justify the use of the proposed data-driven d -selection procedure.
- Relative to the benchmark $\bar{X}$ chart, the d-CCSSBU-L ARL_1 curves generally decline more gradually as k increases. This behavior is expected because the d-CCSSBU-L chart is based on the median rather than the mean. As a more robust measure of central tendency, the median changes less rapidly in response to increasing location shifts than the mean.
- Based on the proposed d -selection procedure, $d = 1$ was selected for Equipment A. Although its ARL_1 curve exhibits the expected gradual decline associated with a median-based monitoring statistic, it achieves a lower ARL_1 value than the benchmark $\bar{X}$ chart at the small shift magnitude $k = 0.5$. Therefore, the selected d-CCSSBU-L chart provides competitive or improved OOC sensitivity while retaining the robustness advantages of a distribution-free monitoring framework.

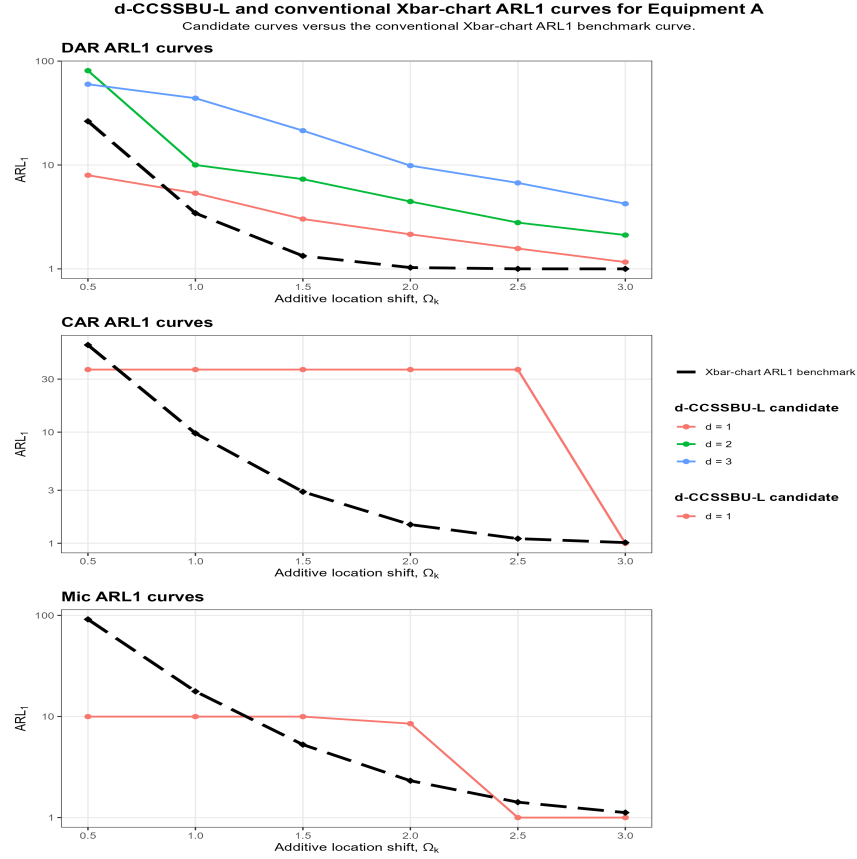


Figure 1: Candidate-specific d-CCSSBU-L ARL1 curves for Equipment A.

3.0.2 d-CCSSBU-L Selection of d and ARL1 Results for Equipment B

Table 2 recommends $d = 2$ for CAR and $d = 1$ for DAR and Mic of Equipment B following Step 8 in ?@sec-d-select-l. The selection_rule column likewise indicates that all recommendations were obtained under Case 2 of Step 8.

Table 2: Recommended d values for the d-CCSSBU-L chart for Equipment B.

Residue	Recommended d	q	Decision case	Observed-behavior case	ARL1 case	A	U	W(d)	G(d)	Selection rule
DAR	1	26	Case 2	D1 intersection D2	XO	{1, 2, 3, 4, 5, 6, 7, 8, 9}	Empty	1.0087	0.4008	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
CAR	2	2	Case 2	D1 intersection D2	XO	{2}	Empty	492.9856	132.8721	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
Mic	1	1	Case 2	D1 intersection D2	XO	{1}	Empty	28.2919	7.6254	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains

Figure 2 visualizes the ARL_1 curves for Equipment B. The same general observations as those for Equipment A apply. An additional observation is that several ARL_1 curves, particularly for CAR and Mic, remain nearly horizontal across the evaluated shift range. This behavior is consistent with the use of the median as the monitoring statistic, which may remain unchanged over a range of shift magnitudes.

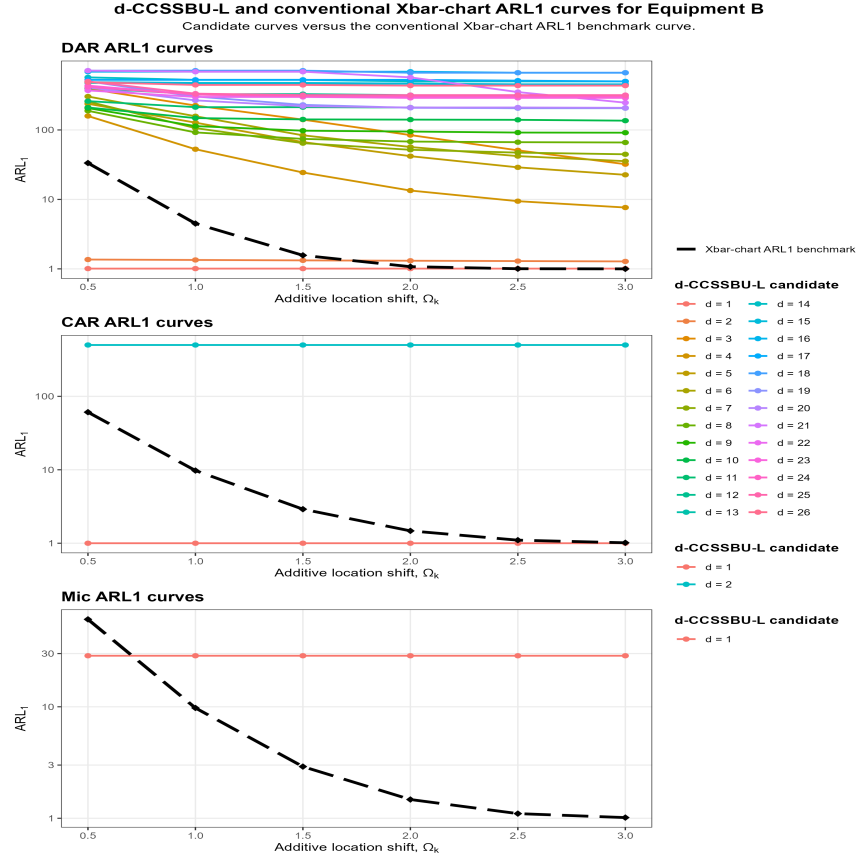


Figure 2: Candidate-specific d-CCSSBU-L ARL1 curves for Equipment B.

Taken together, Figure 1 and Figure 2 show that the optimized d-CCSSBU-L candidates achieved ARL_1 performance comparable to or better than the conventional $\bar{X}$ -chart benchmark, particularly for smaller location shifts. Consequently, ARL_1 performance was not the primary determinant of the final recommended values of d , and the final recommendations were primarily driven by the remaining d-selection criteria described in `?@sec-d-select-l`.

3.1 Selection Results of d for the d-CCSSBU-S Chart

- `d_ccssbu_s()` constructs the d-CCSSBU-S chart for a given value of d which is the first highest risk d clusters used for UCL and CL estimation.
- `make_d_ccssbu_s_arl1_wide_tbl()` is a helper function for `d_ccssbu_s_d_selection_arl1()`.
- `d_ccssbu_s_d_selection_arl1()` applies `d_ccssbu_s()` across all candidate values $d = 1, \dots, q$ using Excel input data and summarizes OOC counts and ARL_1 performance for DAR, CAR, and Mic residue types and use `make_d_ccssbu_s_arl1_wide_tbl()` to make an output table.
- `render_d_ccssbu_s_arl1_tbl()` renders the output table from `make_d_ccssbu_s_arl1_wide_tbl()` as a responsive tibble that automatically adjusts to page width.

- `render_d_ccssbu_s_recommendation_tbl()` generates the recommended d selection table for each residue type (DAR, CAR, and Mic).
- `plot_d_ccssbu_s_arl1_curves()` plot all ARL_1 as y-axis values and grid of k as x-axis for all d-CCSSBU-S charts and X-bar chart.

3.1.1 d-CCSSBU-S Selection of d and ARL_1 Results for Equipment A

Table 3 recommends $d = 1$ for DAR, CAR, and Mic of Equipment A following Step 8 in ?@sec-d-select-s. The `selection_rule` column indicates that all recommendations were obtained under Case 2 of Step 8.

Table 3: Recommended d values for the d-CCSSBU-S chart for Equipment A.

Residue	d	q	Case	Observed Case	ARL Case	A	U	W(d)	G(d)	Rule
CAR	1	1	Case 2	D1_intersect_D2	XO	1		4.5574	2.6012	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
DAR	1	3	Case 2	D1_intersect_D2	XO	1, 2		4.4881	1.5006	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
Mic	1	1	Case 2	D1_intersect_D2	XO	1		1.5395	0.1828	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains

Figure 3 visualizes the corresponding ARL_1 curves for the d-CCSSBU-S chart for Equipment A. Similar observations to those for the d-CCSSBU-L chart apply. One notable difference is that the relative performance of the d-CCSSBU-S and conventional S charts varies by residue type. For DAR and Mic, the d-CCSSBU-S candidates achieve ARL_1 values comparable to or lower than those of the benchmark S chart across portions of the evaluated shift range, whereas for CAR the conventional S chart consistently exhibits lower ARL_1 values, indicating greater sensitivity to variability shifts. Following the proposed d-selection procedure, $d = 1$ was selected for DAR, CAR, and Mic because it achieved the smallest ARL_1 among the admissible candidates.

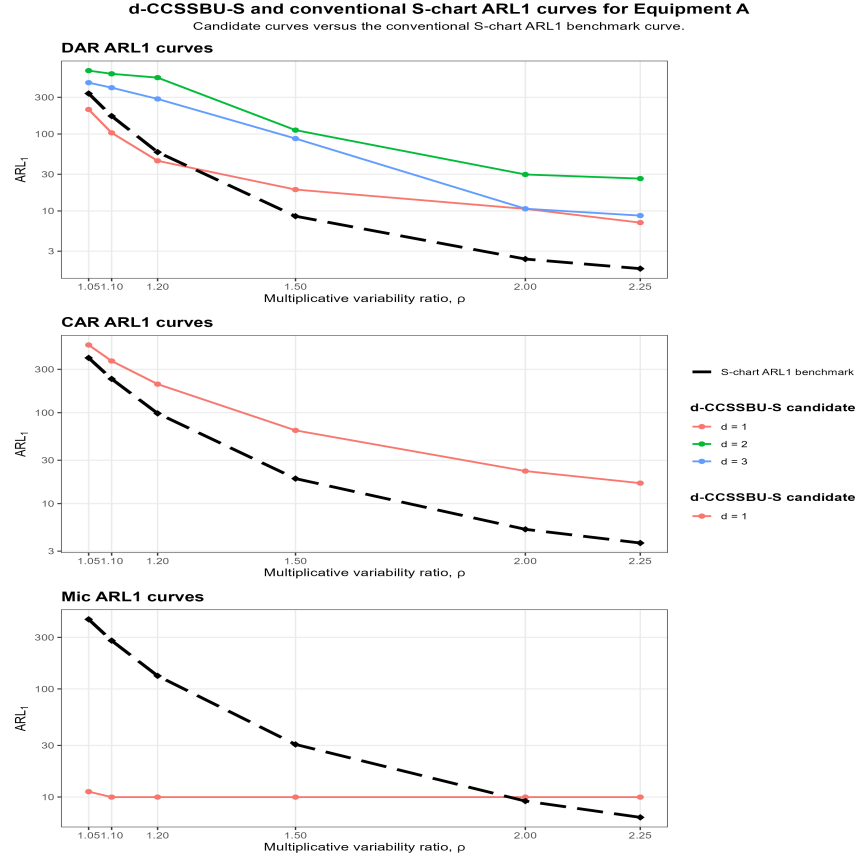


Figure 3: Candidate-specific d-CCSSBU-S ARL₁ curves for Equipment A.

3.1.2 d-CCSSBU-S Selection of d and ARL₁ Results for Equipment B

Table 4 recommends $d = 13$ for DAR and $d = 1$ for CAR and Mic of Equipment B following Step 8 in ?@sec-d-select-s. The selection_rule column indicates that all recommendations were obtained under Case 2 of Step 8.

Table 4: Recommended d values for the d-CCSSBU-S chart for Equipment B.

Residue	d	q	Case	Observed Case	ARL Case	A	U	W(d)	G(d)	Rule
CAR	1	2	Case 2	D1_intersect_D2	XO	1, 2		1.5221	1.1434	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
DAR	13	26	Case 4	D1	XO	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26		7.3323	3.0336	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains
Mic	1	1	Case 2	D1_intersect_D2	XO	1		35.9163	3.4234	XO case: minimum W over A; minimum G among candidates tied on W; maximum d if a tie remains

Figure 4 visualizes the corresponding ARL₁ results for Equipment B. The overall ARL₁ patterns are similar to those observed for the d-CCSSBU-L charts. Consistent with the d-CCSSBU-L results, the selected d made the d-CCSSBU-S chart have lower ARL₁ values than the conventional S -chart benchmark for smaller variability shifts, indicating improved sensitivity to the early detection of variability changes.

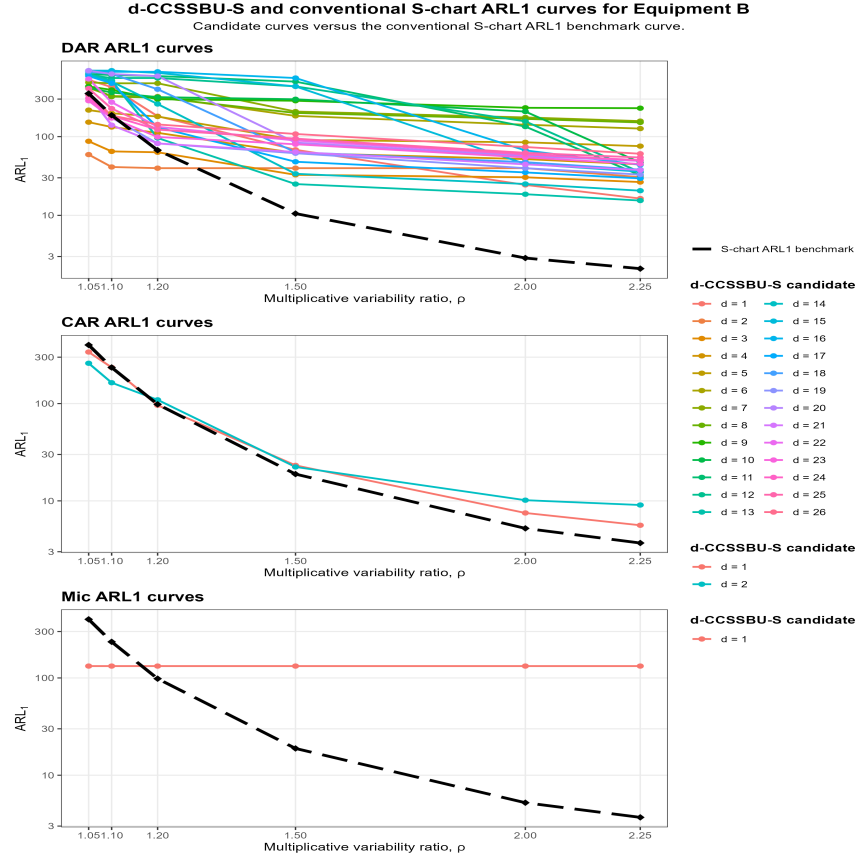


Figure 4: Candidate-specific d-CCSSBU-S ARL₁ curves for Equipment B.

Figure 3 and Figure 4 visualize the ARL₁ performance of the candidate d-CCSSBU-S charts relative to the conventional *S*-chart benchmark. Across both equipment datasets, the optimized d-CCSSBU-S candidates generally achieved smaller ARL₁ values than the benchmark *S* chart for smaller variability shifts, indicating improved OOC sensitivity. Although the conventional *S* chart exhibited lower ARL₁ values than the d-CCSSBU-S candidates for CAR in Equipment A, the overall ARL₁ performance of the optimized d-CCSSBU-S charts remained comparable to or better than that of the benchmark. Similar to the d-CCSSBU-L results, the d-CCSSBU-S ARL₁ curves were generally less responsive to increasing variability shifts than the conventional *S*-chart curve.

Therefore, ARL₁ performance remained acceptable relative to the benchmark and did not prevent selection of the recommended charts. The final recommendations were determined by the complete d-selection procedure described in `?@sec-d-select-s`, which jointly considered calibration quality, ND-event behavior, Stage 3A performance, and ARL₁ performance.

3.2 d-CCSSBU Charts for Recommended d

The d-CCSSBU-L and d-CCSSBU-S charts for DAR are presented using the recommended values of d for Equipment A (representing Stage 3A) and Equipment B (representing Stage 3B). Similar

to the conventional $\bar{X}$ -chart and S -chart pair, the d-CCSSBU-L chart monitors process location, whereas the d-CCSSBU-S chart monitors within-event variability. Together, the two charts provide complementary assessments of multicluster process stability.

3.2.1 d-CCSSBU Charts for DAR of Equipment A and B

Figure 5 presents the d-CCSSBU-L and d-CCSSBU-S charts for DAR using the recommended values of d . For Equipment A, no OOC signals were observed on either the d-CCSSBU-L or d-CCSSBU-S chart, indicating stable process location and variability. For Equipment B, no OOC signals were observed on the d-CCSSBU-L chart, indicating stable process location performance. On the d-CCSSBU-S chart, Event 10 was identified as OOC. Inspection of the corresponding DAR observations (15.93, 0.11, 0.11, 15.93, and 32.95) shows substantial within-event variability, resulting in a large MAD value.

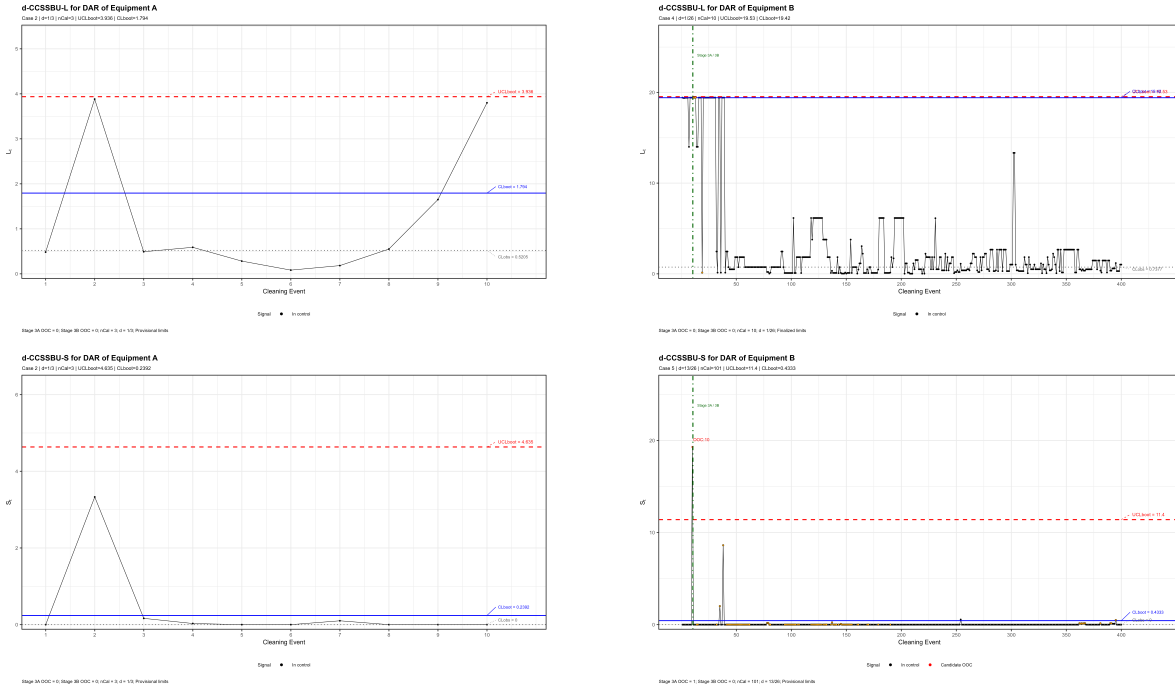


Figure 5: d-CCSSBU-L and d-CCSSBU-S charts for DAR of Equipment A and B.

3.2.2 d-CCSSBU Charts for CAR of Equipment A and B

Figure 6 presents the d-CCSSBU-L and d-CCSSBU-S charts for CAR using the recommended values of d . For Equipment A, no OOC signals were observed on either the d-CCSSBU-L or d-CCSSBU-S chart, indicating stable process location and within-event variability.

For Equipment B, Event 19 was identified as OOC on the d-CCSSBU-L chart but not on the d-CCSSBU-S chart. The corresponding CAR observations were 35.49, 31.99, and 24.80, yielding a

median value of 31.99 that was substantially higher than those of the surrounding events. In contrast, the within-event variability remained relatively small, resulting in an in-control d-CCSSBU-S statistic. Consequently, Event 19 was detected as a location shift rather than a scale shift.

Equipment B contains CAR data from both the original cleaning process and its improved cleaning process using a new cleaning agent. By treating the two cleaning processes as separate clusters, the d-CCSSBU framework enables their performance to be evaluated within a common monitoring system. The resulting charts show that the improved cleaning process remains comparable to the original process and can be monitored using common control limits. This example illustrates that multicluster SPC can be used not only for stability monitoring but also for comparing related cleaning processes while accounting for their inherent differences.

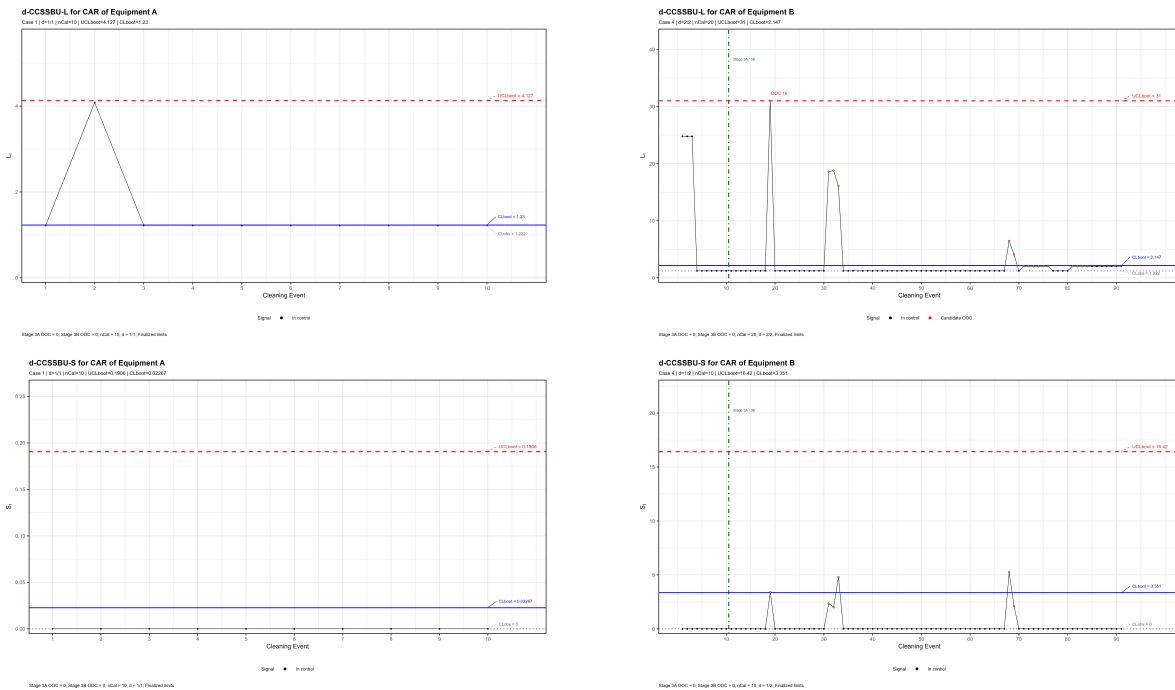


Figure 6: d-CCSSBU-L and d-CCSSBU-S charts for CAR of Equipment A and B.

3.2.3 d-CCSSBU Charts for Mic of Equipment A and B

Figure 7 presents the d-CCSSBU-L and d-CCSSBU-S charts for Mic using the recommended values of d . For both Equipment A and Equipment B, no OOC signals were observed on either the d-CCSSBU-L or d-CCSSBU-S chart, indicating stable microbial process performance.

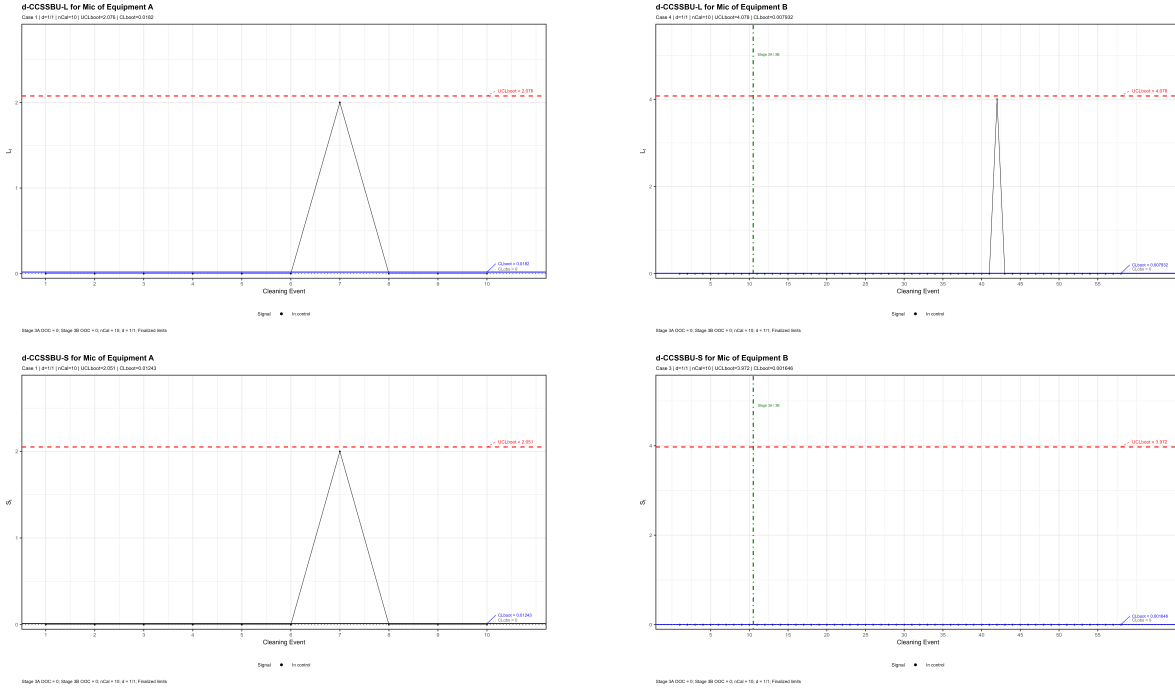


Figure 7: d-CCSSBU-L and d-CCSSBU-S charts for Mic of Equipment A and B.

3.3 Poisson u Chart Illustration for Mic Data

3.3.1 Poisson u Chart for Mic Data of Equipment A

Table 5 summarizes the Poisson goodness-of-fit, overdispersion, and chart-selection results for Mic data from Equipment A and indicates that a conventional Poisson u chart is appropriate for this dataset.

Table 5: Mic data Poisson assessment and chart selection for Equipment A.

Equipment	Case	USL	GOF p	GOF pass	Dispersion	Overdispersion pass	Mic Poisson	CL	UCL min	UCL max	Stage 3A OOC	Stage 3B OOC
Equipment A	Case 3	25	0.627	TRUE	1	TRUE	TRUE	0.2	2.0974	2.0974	0	0

Figure Figure 8 shows that no OOC signals were observed on the Poisson u chart, indicating stable microbial process performance.

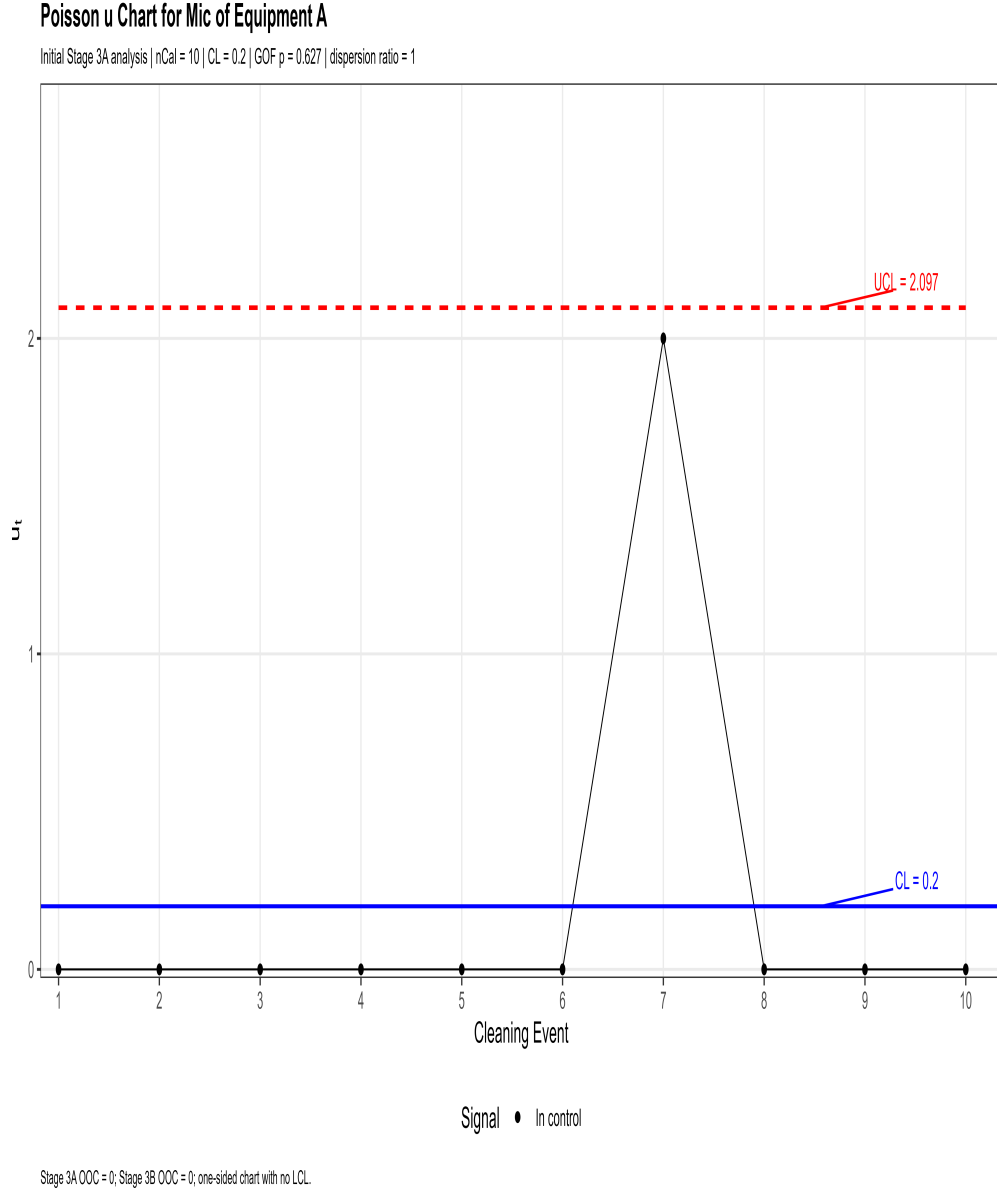


Figure 8: Poisson u chart for Mic data from Equipment A.

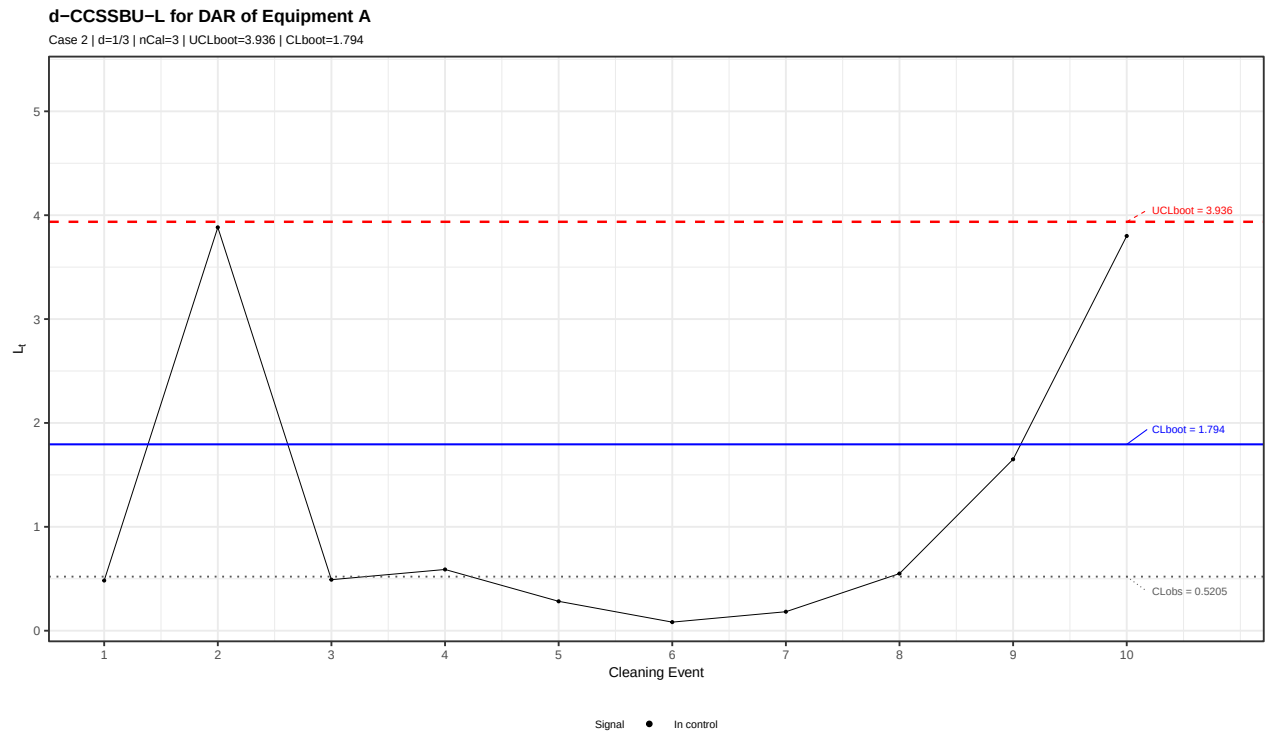
The Poisson u chart in Figure 8 is nearly identical to the d-CCSSBU-L and d-CCSSBU-S charts in Figure 7. This similarity arises from the structure of the data. Each cleaning event contains two observations, for which the event mean and median are identical. Moreover, the only event with nonzero microbial counts (Event 7) has observations of 0 and 1, yielding a mean, median, and MAD of 0.5. After USL-normalization, both the location statistic L_t and scale statistic S_t take the same value of 2. The corresponding UCLs are also very similar (2.051, 2.076, and 2.097), differing by less than 3%. As a result, all three charts yield the same stability assessment for this dataset. This agreement suggests that the distribution-free d-CCSSBU framework and the parametric Poisson u chart provide consistent monitoring conclusions for Equipment A.

3.3.2 Non-Poisson Mic Data of Equipment B

Table 6 shows the Mic data for Equipment B do not satisfy the Poisson assumptions. The goodness-of-fit test failed ($p = 0.0274$), and the overdispersion assessment also failed with a dispersion ratio of 1.6286. Consequently, the Mic data were not considered Poisson, and a conventional Poisson u chart was not appropriate. This conclusion is also supported by the d-CCSSBU charts. Unlike Equipment A, where the d-CCSSBU-L and d-CCSSBU-S charts were nearly identical and closely matched the Poisson u chart, the location and scale charts for Equipment B exhibit different patterns. The visual differences between the two charts indicate that the underlying process behavior is not consistent with a simple Poisson mechanism.

Table 6: Mic data Poisson assessment and chart selection for Equipment B.

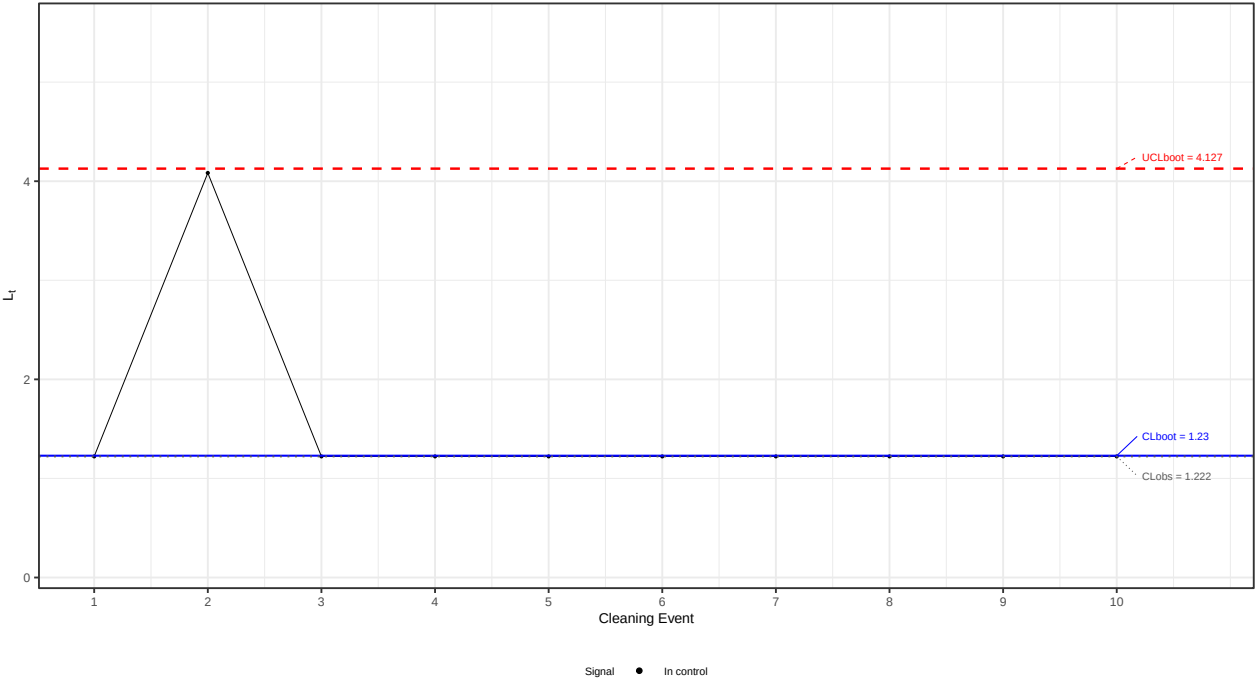
Equipment	Case	USL	GOF p	GOF pass	Dispersion	Overdispersion pass	Mic Poisson	CL	UCL min	UCL max	Stage 3A OOC	Stage 3B OOC
Equipment B	Case 3	25	0.0274	FALSE	1.6286	FALSE	FALSE	0.3333	1.488	2.3333	0	0



Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 3; d = 1/3; Provisional limits

d-CCSSBU-L for CAR of Equipment A

Case 1 | $d=1/1$ | $nCal=10$ | $UCLboot=4.127$ | $CLboot=1.23$



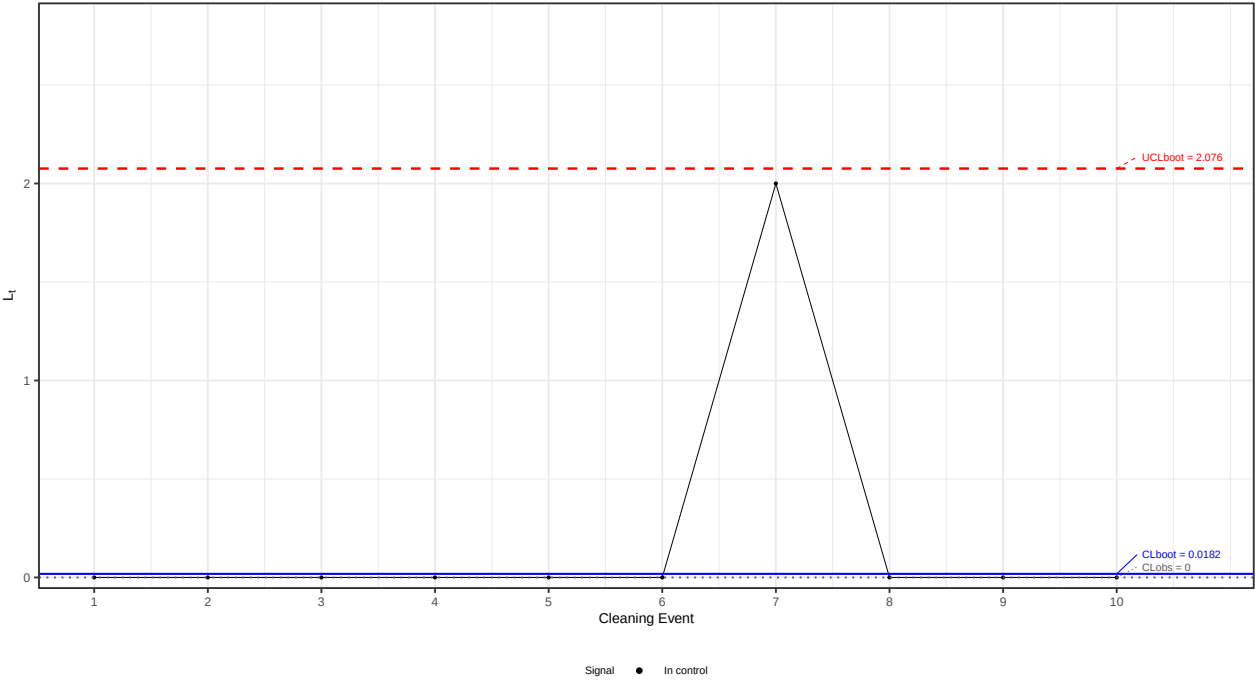
168

Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 10$; $d = 1/1$; Finalized limits

169

d-CCSSBU-L for Mic of Equipment A

Case 1 | $d=1/1$ | $nCal=10$ | $UCLboot=2.076$ | $CLboot=0.0182$



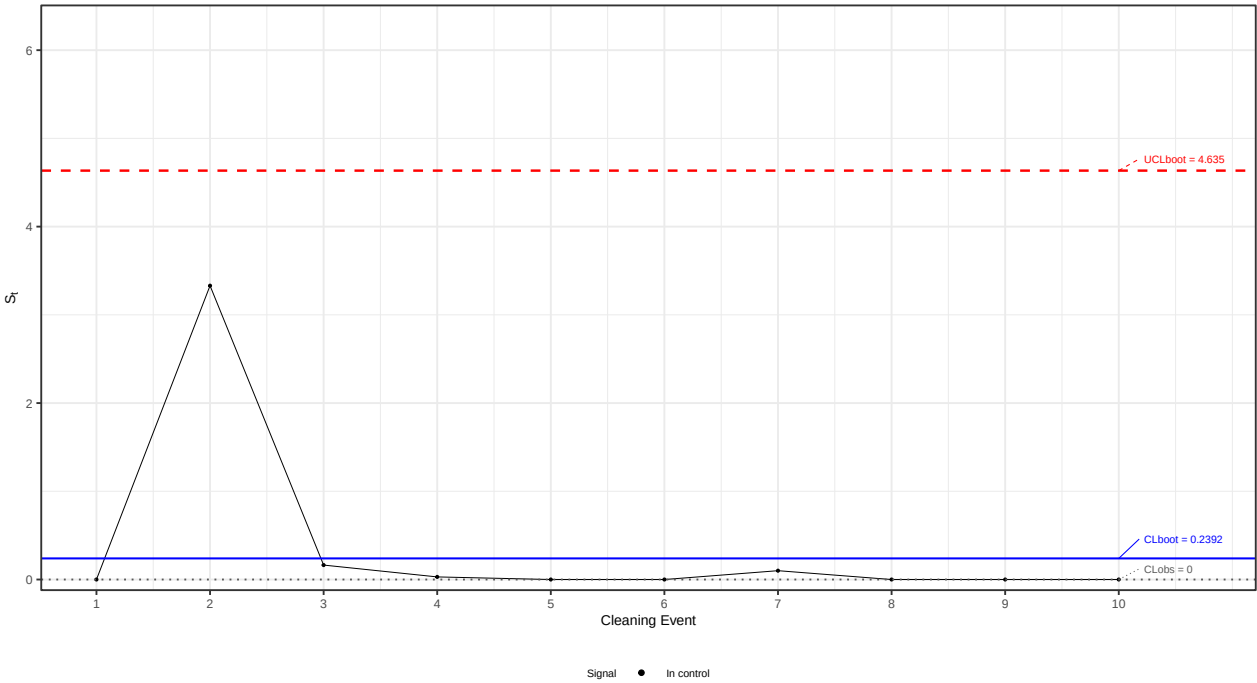
170

Stage 3A OOC = 0; Stage 3B OOC = 0; $nCal = 10$; $d = 1/1$; Finalized limits

171

d-CCSSBU-S for DAR of Equipment A

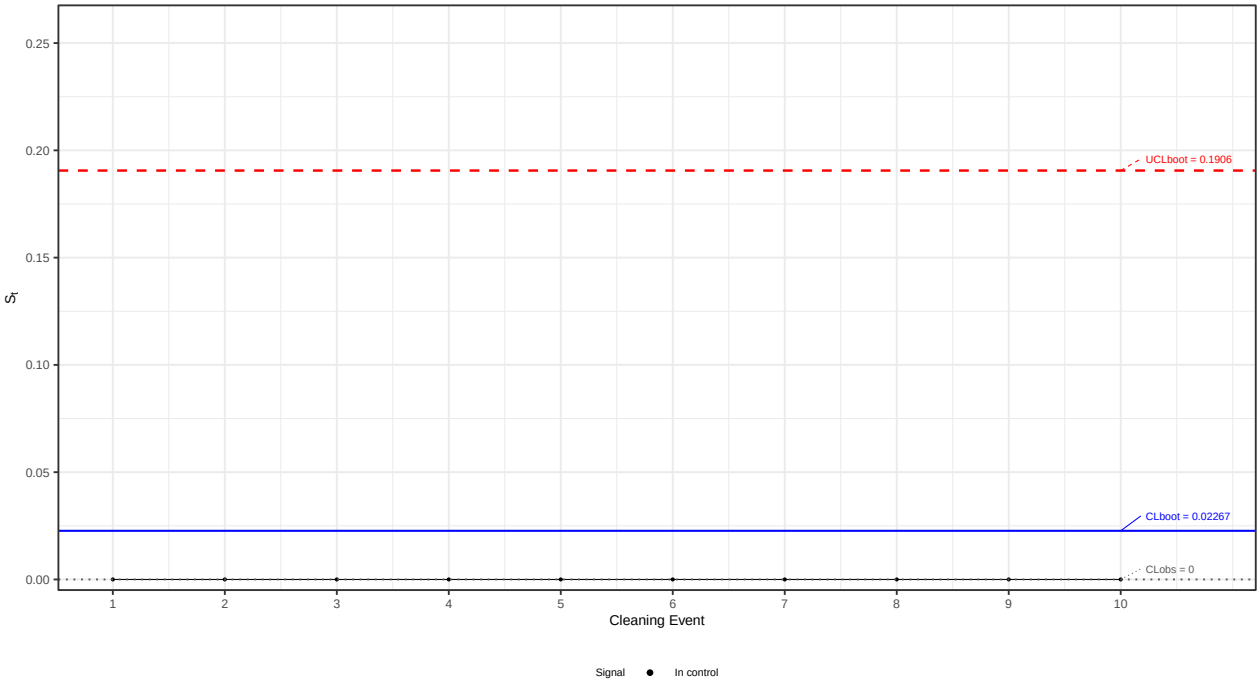
Case 2 | d=1/3 | nCal=3 | UCLboot=4.635 | CLboot=0.2392



172 Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 3; d = 1/3; Provisional limits

d-CCSSBU-S for CAR of Equipment A

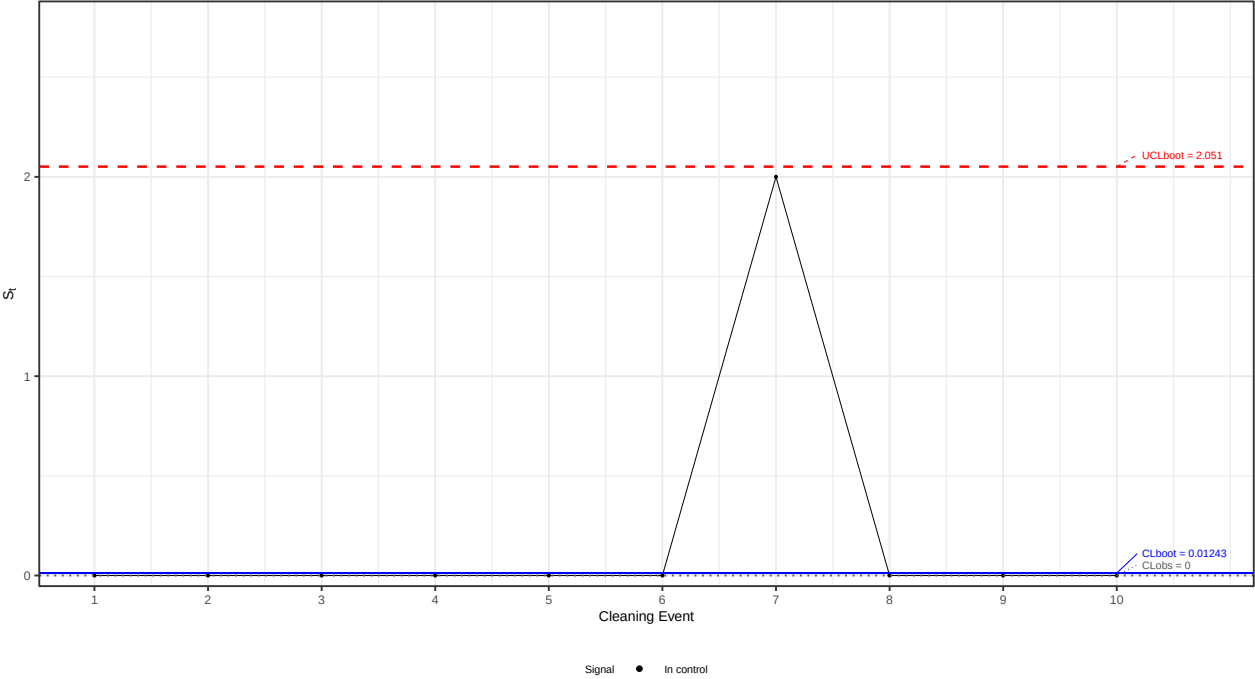
Case 1 | d=1/1 | nCal=10 | UCLboot=0.1906 | CLboot=0.02267



174 Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 10; d = 1/1; Finalized limits

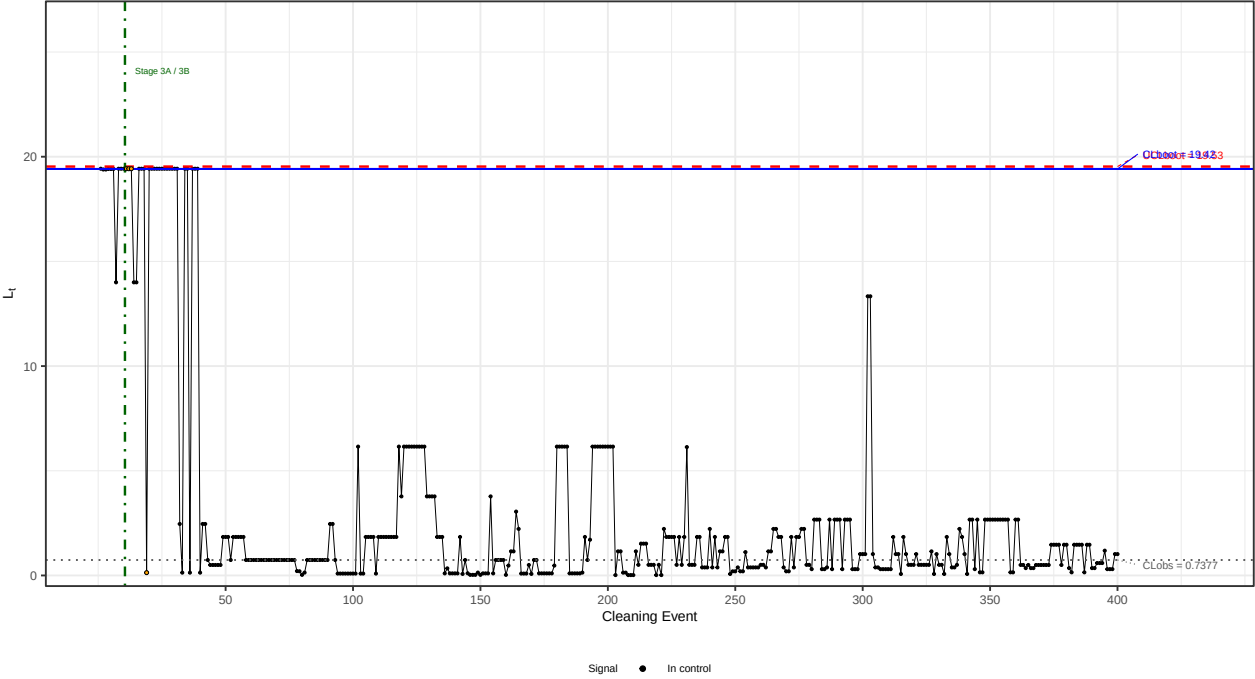
175

d-CCSSBU-S for Mic of Equipment A
Case 1 | d=1/1 | nCal=10 | UCLboot=2.051 | CLboot=0.01243



176 Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 10; d = 1/1; Finalized limits

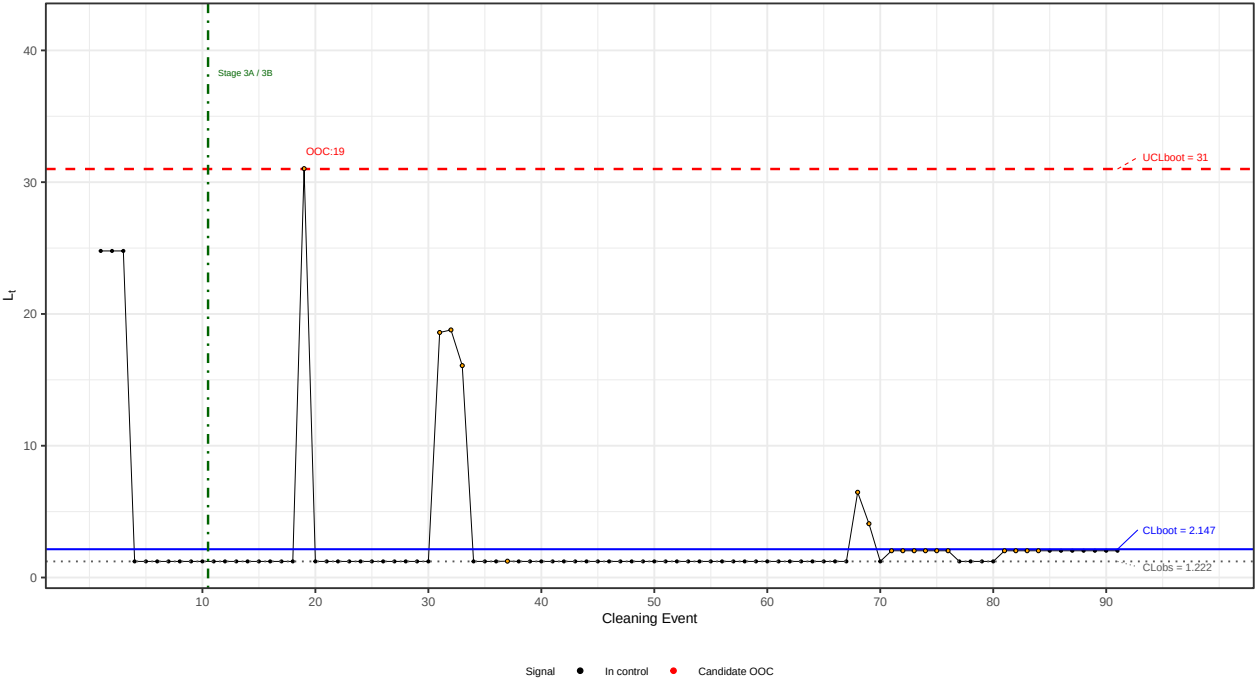
d-CCSSBU-L for DAR of Equipment B
Case 4 | d=1/26 | nCal=10 | UCLboot=19.53 | CLboot=19.42



178 Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 10; d = 1/26; Finalized limits

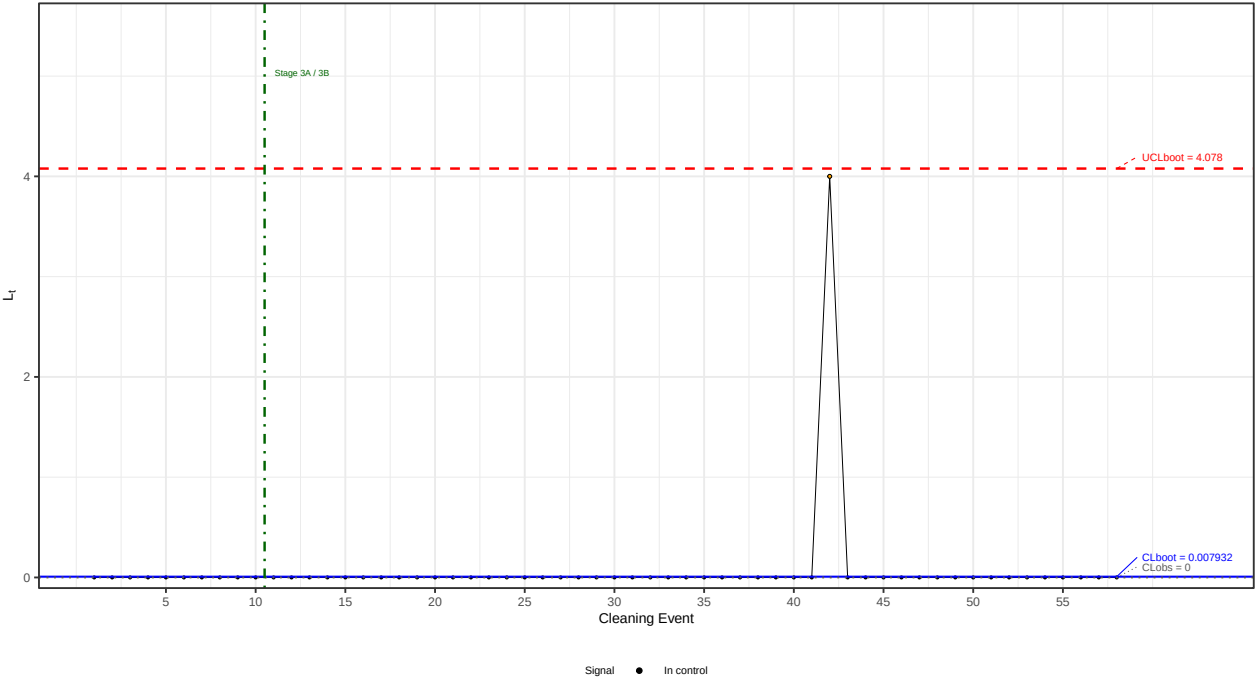
179

d-CCSSBU-L for CAR of Equipment B
Case 4 | d=2/2 | nCal=20 | UCLboot=31 | CLboot=2.147



180 Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 20; d = 2/2; Finalized limits

d-CCSSBU-L for Mic of Equipment B
Case 4 | d=1/1 | nCal=10 | UCLboot=4.078 | CLboot=0.007932

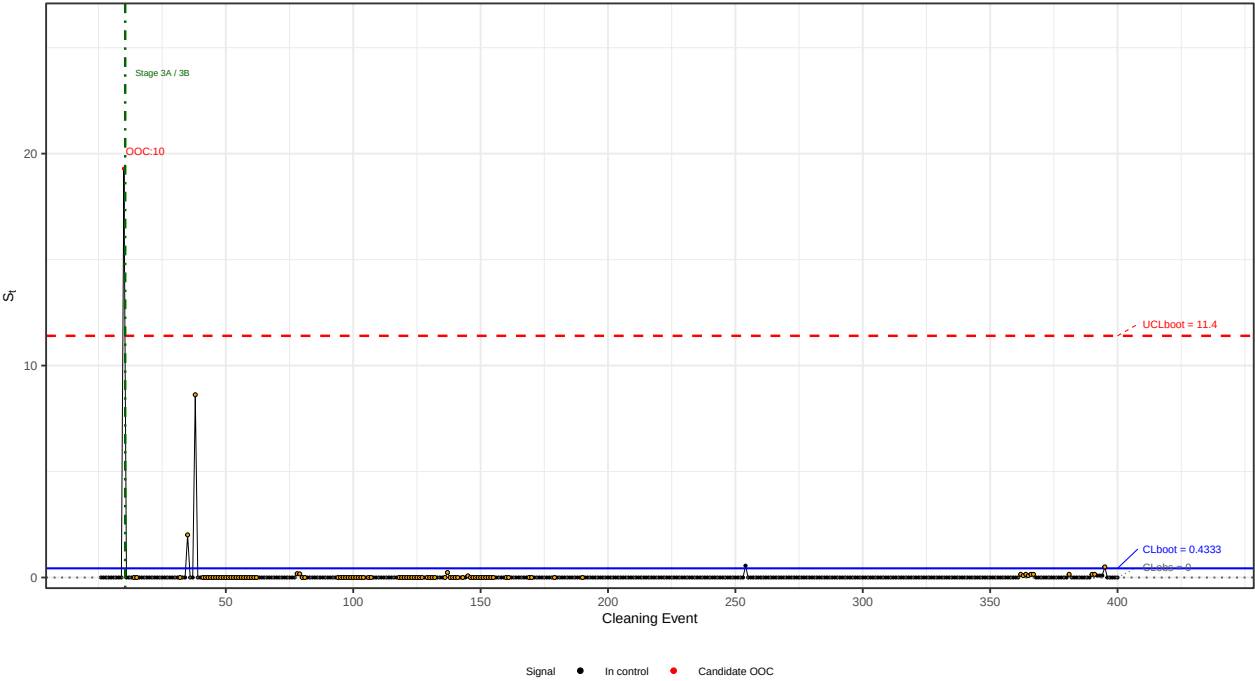


182 Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 10; d = 1/1; Finalized limits

183

d-CCSSBU-S for DAR of Equipment B

Case 5 | d=13/26 | nCal=101 | UCLboot=11.4 | CLboot=0.4333



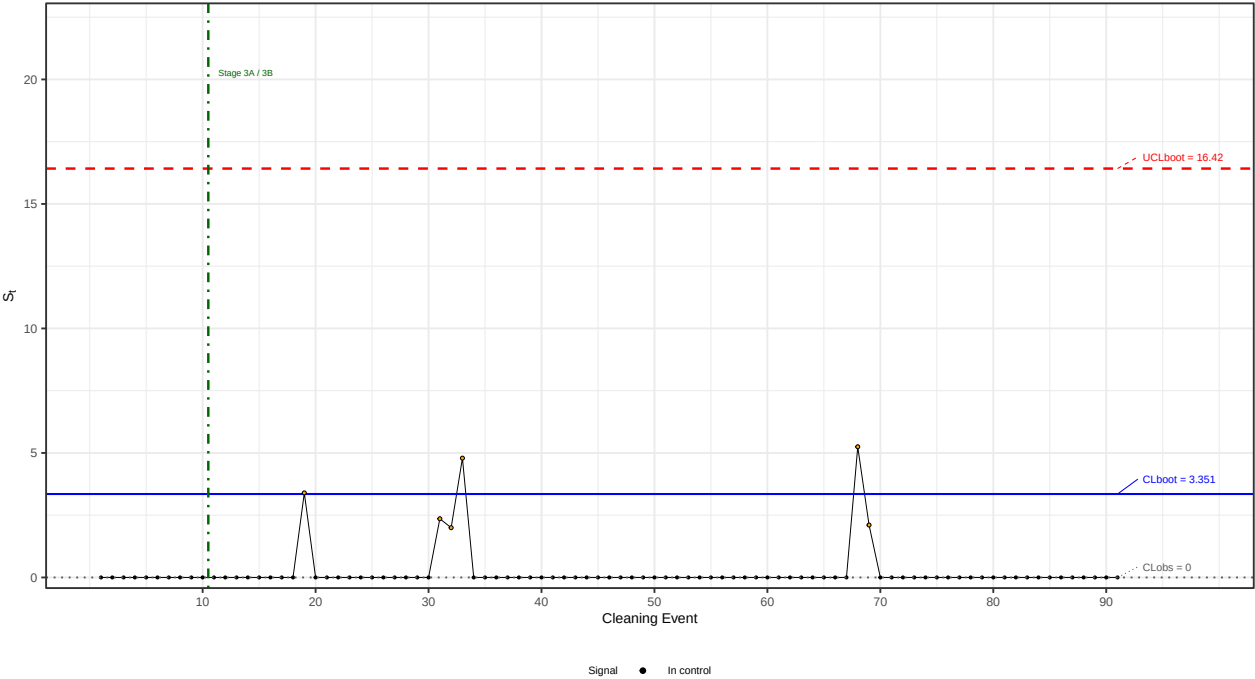
184

Stage 3A OOC = 1; Stage 3B OOC = 0; nCal = 101; d = 13/26; Provisional limits

185

d-CCSSBU-S for CAR of Equipment B

Case 4 | d=1/2 | nCal=10 | UCLboot=16.42 | CLboot=3.351



186

Stage 3A OOC = 0; Stage 3B OOC = 0; nCal = 10; d = 1/2; Finalized limits

187

d-CCSSBU-S for Mic of Equipment B

Case 3 | d=1/1 | nCal=10 | UCLboot=3.972 | CLboot=0.001646

